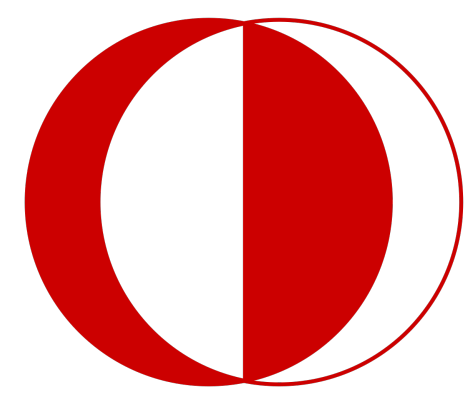




AUTOMATED CONSTRUCTION of MULTI-ELEMENT MEAM INTERATOMIC POTENTIALS

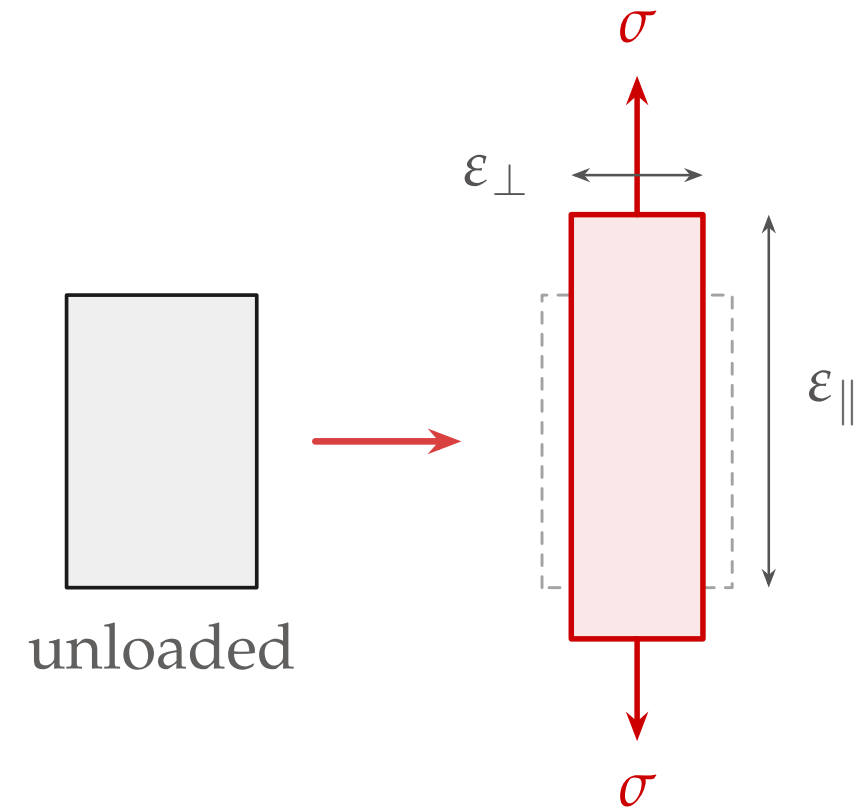
A DFT-informed neural-network optimization framework

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1 Motivation & background

Two numbers govern how a structural alloy carries load — and a neural network can predict both from composition alone.



- **Young's modulus** $E = \sigma / \epsilon_{\parallel}$ — stiffness, the stress needed to stretch it (GPa).
- **Poisson's ratio** $\nu = -\epsilon_{\perp} / \epsilon_{\parallel}$ — how much it thins sideways when pulled.

For a cubic crystal both reduce to two stiffness constants — C_{11} (axial) and C_{12} (transverse):

$$E = \frac{(C_{11} - C_{12})(C_{11} + 2C_{12})}{C_{11} + C_{12}}, \quad \nu = \frac{C_{12}}{C_{11} + C_{12}}, \quad B = \frac{1}{3}(C_{11} + 2C_{12}).$$

- Stiff alloy (E large) resists stretching; low ν barely thins under load.
- E and ν fix deflection, vibration and buckling — the first properties screened for a structural alloy, and the targets this pipeline predicts and optimizes.

THE BOTTLENECK

- Molecular dynamics recovers E and ν cheaply — but only with an accurate interatomic potential.
- The standard metallic form, the Modified Embedded Atom Method (MEAM) [1, 2], is fitted *by hand*.
- Hand fitting is expert work that does not scale past a few elements.

Objective. Build MEAM potentials *automatically* for any set of elements — no hand fitting, no experimental input.

THE DESIGN SPACE

- Compositions drawn on the 12-element simplex (Al, Co, Cr, Cu, Fe, Mg, Mn, Mo, Ni, Si, Ti, Zn), weighted toward aluminium.
- Of ~ 7900 random configurations, **7815** survive the physicality filter ($E > 0$, $0 < \nu < 0.48$, $C_{11} > C_{12}$).

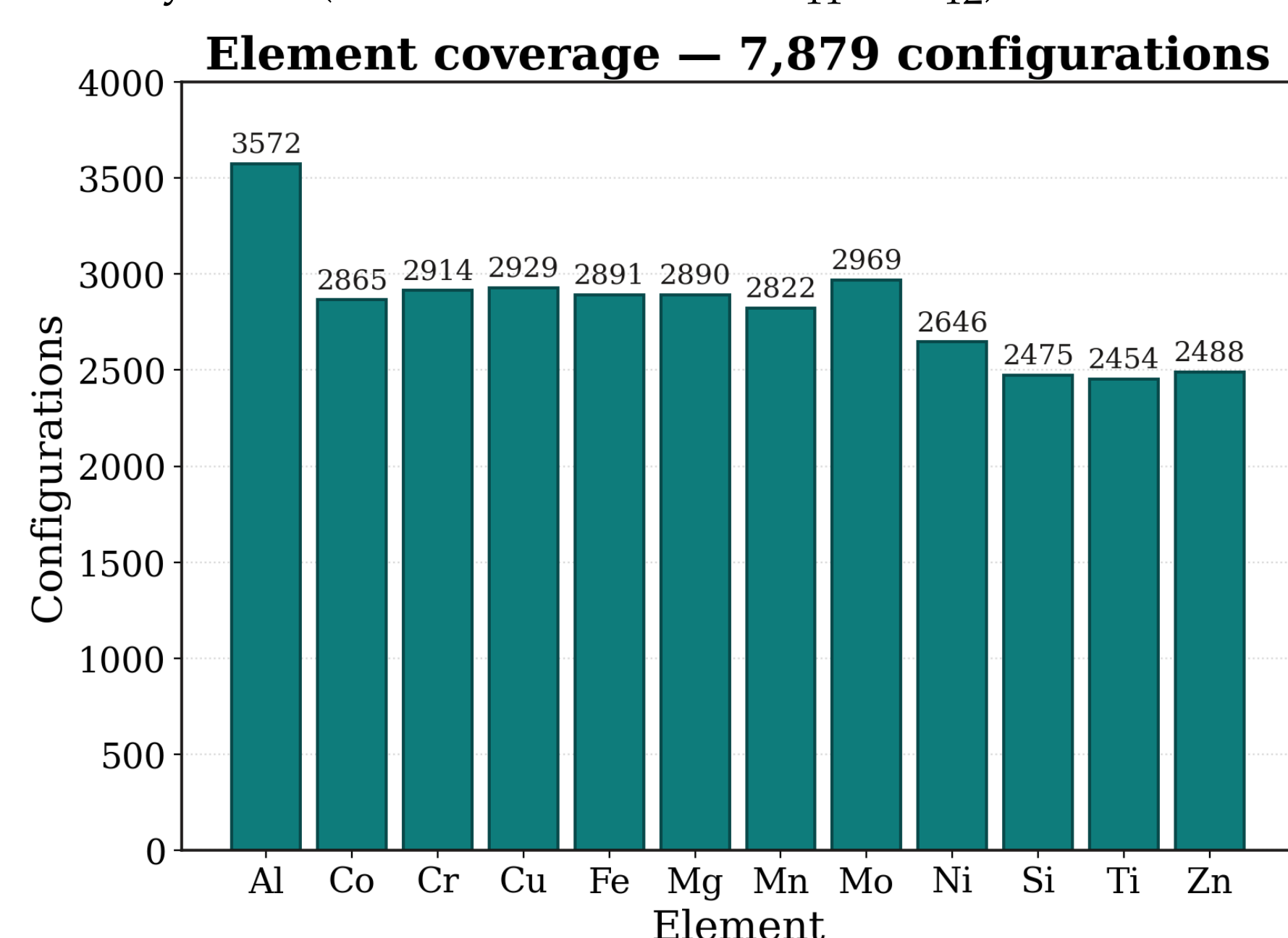


Fig. 1. Element coverage across the retained corpus.

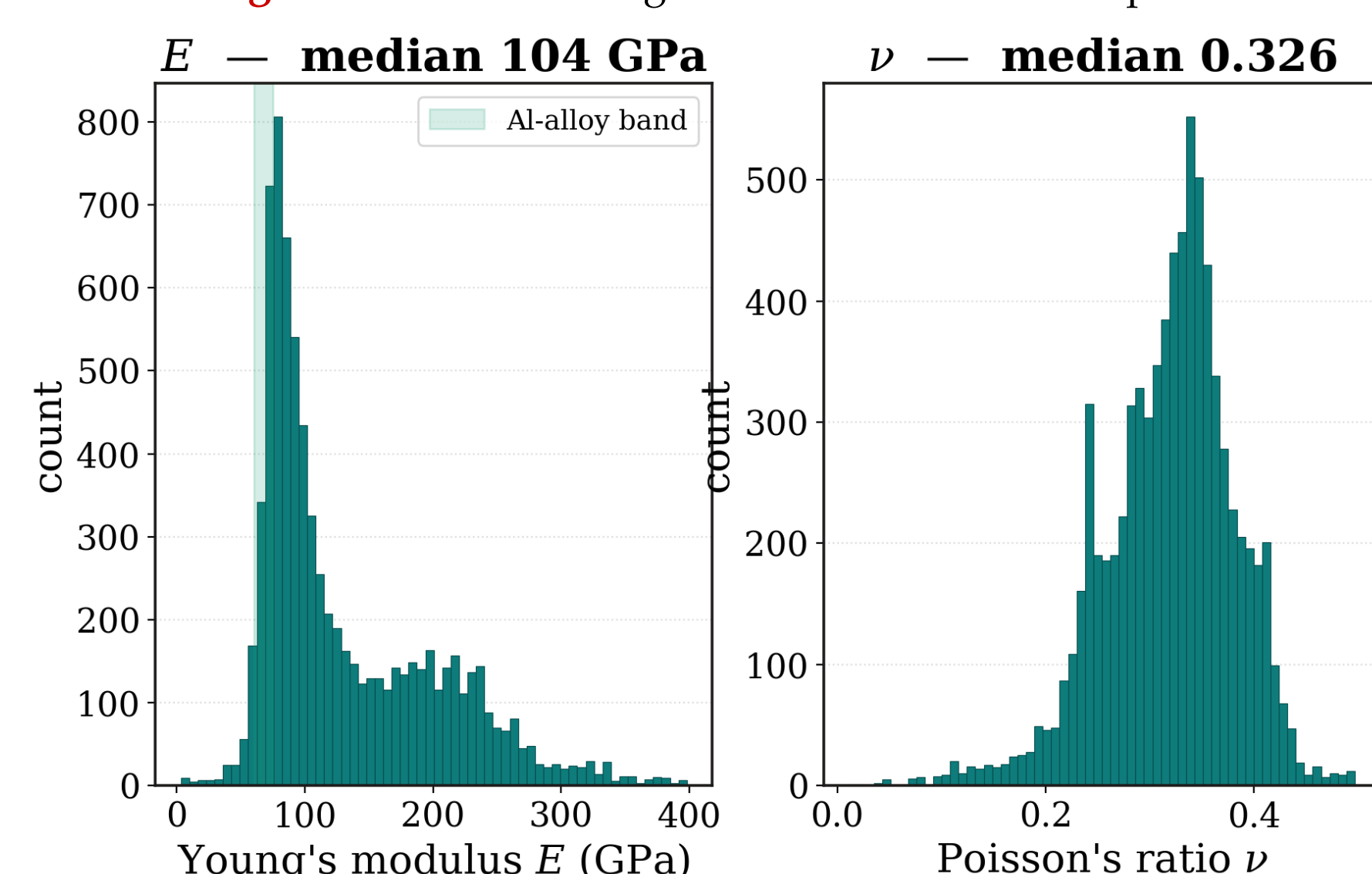


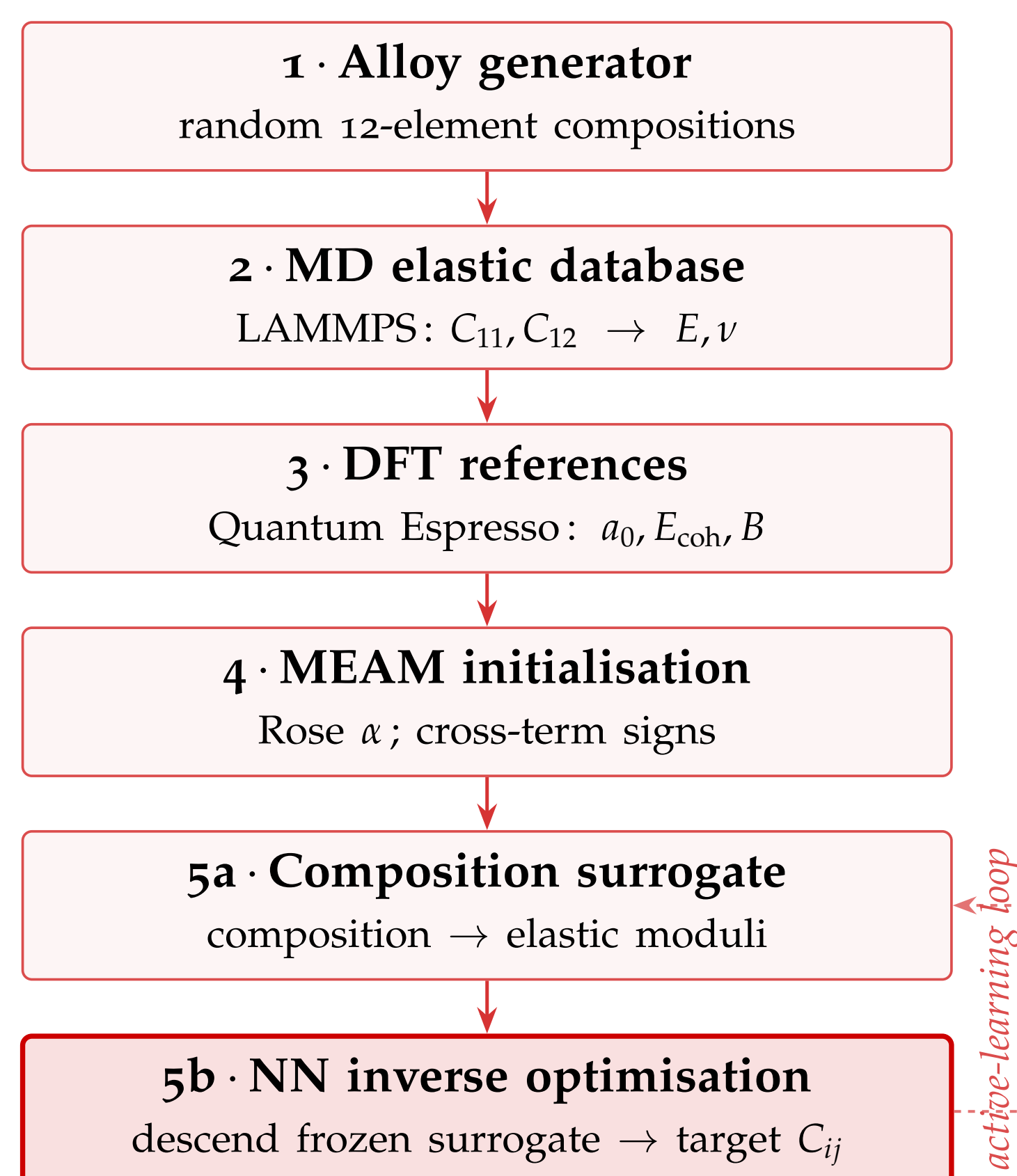
Fig. 2. Property space: E and ν over the 7815-alloy corpus.

2 Methodology

An automated five-stage pipeline turns random compositions into an optimized multi-element potential.

Three engines at a glance

- **DFT** (Quantum Espresso) — first-principles a_0 , E_{coh} , B and binary formation energies: the ground-truth references.
- **MEAM** — many-body metallic potential (Rose binding curve + cross-terms), DFT-seeded and neural-network refit.
- **MD** (LAMMPS) — strain sweep on each potential $\rightarrow C_{11}, C_{12} \rightarrow E, \nu$: the elastic database.



DFT-SEEDING INITIALISATION

- Quantum Espresso [4] fixes each element's a_0 , E_{coh} and bulk modulus B , pinning the Rose binding curve [5] and the MEAM decay exponent

$$\alpha = \sqrt{9B\Omega/E_{\text{coh}}}$$

(Ω the atomic volume; $\alpha \approx 4.5$ for Al, up to 10.9 for Zn).

- Sixty-six binary formation energies seed each cross-term's sign.

	DFT binary formation energy E_{form} (eV/atom)												
Mg	+0.88	+0.81	+0.41	+0.27	+0.55	-0.06	+0.09	+0.49	+0.08	+0.05	+1.39		
Mo	+0.88		+1.21	+1.19	+0.68	+0.09	+4.05	-0.35	-0.93	-0.85	-0.68		+0.02
Al	+0.81	+1.21		+0.03	+1.82	+0.68	+0.83	+1.36	+0.60	+0.75	+0.42		-0.16
Cu	+0.41	+1.19	+0.03		+1.72	+1.51	+1.11	+2.15	+0.65	+0.67	+0.33		-0.07
Zn	+0.27	+0.68	+1.82	+1.72		+0.58	+0.47	+0.84	+0.59	+0.20	+0.23		+1.67
Ti	+0.55	+0.09	+0.68	+1.51	+0.58		+0.21	+0.19	+0.11	-0.47	-0.34		+0.52
Si	-0.06	+4.05	+0.83	+1.11	+0.47	+0.21		+0.47	+2.25	+2.71	+2.05		+1.47
Co	+0.09	-0.35	+1.36	+2.15	+0.84	+0.19	+0.47		+0.36	+0.20	+0.17		+2.33
Cr	+0.49	-0.93	+0.60	+0.65	+0.59	+0.11	+2.25	+0.36		+0.17	+0.01		+0.66
Fe	+0.08	-0.85	+0.75	+0.67	+0.20	-0.47	+2.71	+0.20	+0.17		+0.05		+0.59
Mn	+0.05	-0.68	+0.42	+0.33	+0.23	-0.34	+2.05	+0.17	+0.01	+0.05			+0.23
Ni	+1.39	+0.02	-0.16	-0.07	+1.67	+0.52	+1.47	+2.33	+0.66	+0.59	+0.23		
	Mg	Mo	Al	Cu	Zn	Ti	Si	Co	Cr	Fe	Mn	Ni	

Fig. 3. DFT binary formation energies (eV/atom); negative = exothermic.

NEURAL SURROGATE & INVERSE DESIGN

- A fully connected network maps composition to the stiffness constants (E, ν follow analytically).
- A second network learns (MEAM parameters) $\rightarrow (C_{11}, C_{12})$.
- Gradient descent through the *frozen* surrogate then locates parameters that hit a target.

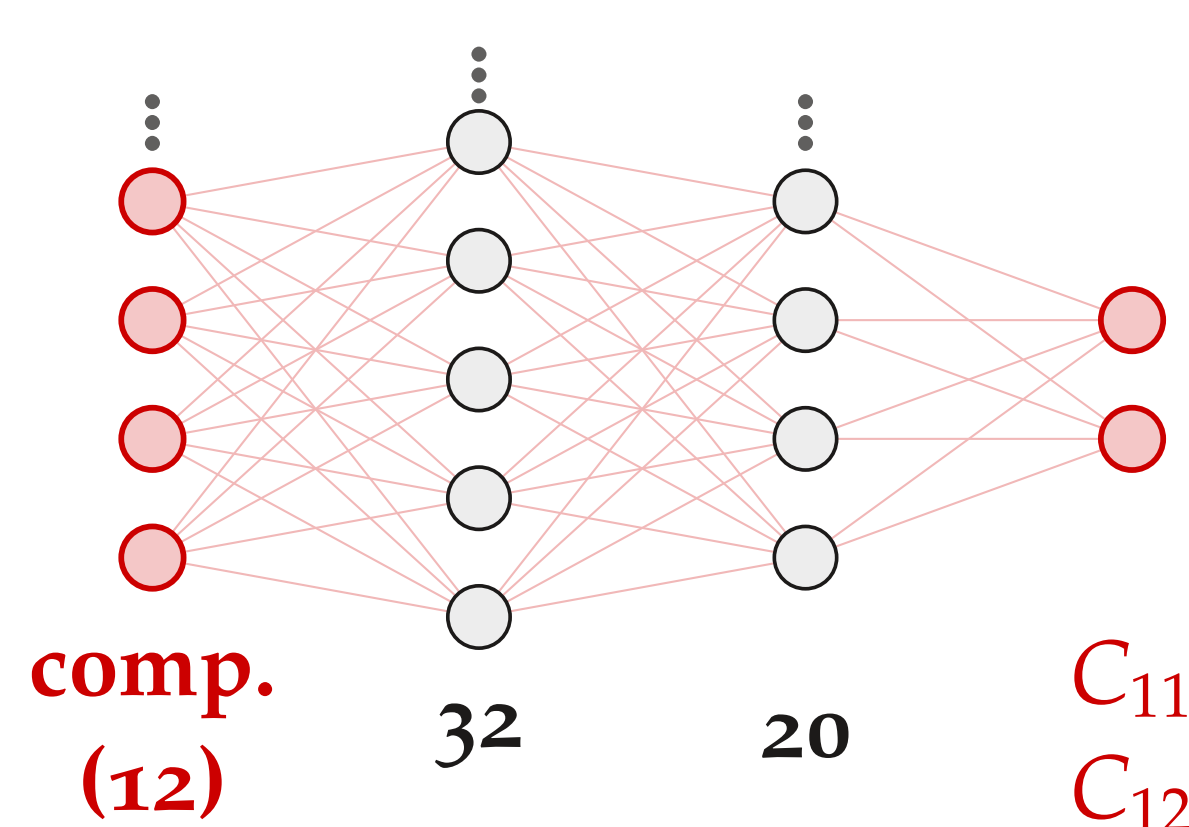


Fig. 4. Surrogate architecture: composition $\rightarrow$ two ReLU hidden layers $\rightarrow (C_{11}, C_{12})$.

3 Results & discussion

ELASTIC DATABASE

- ~ 4 nm LAMMPS [3] supercell per alloy; axial strains ($\delta = 10^{-3}$) give C_{11}, C_{12} and hence E and $\nu = C_{12}/(C_{11} + C_{12})$.
- Median $E = 103.6$ GPa, $\nu = 0.326$ over the corpus.

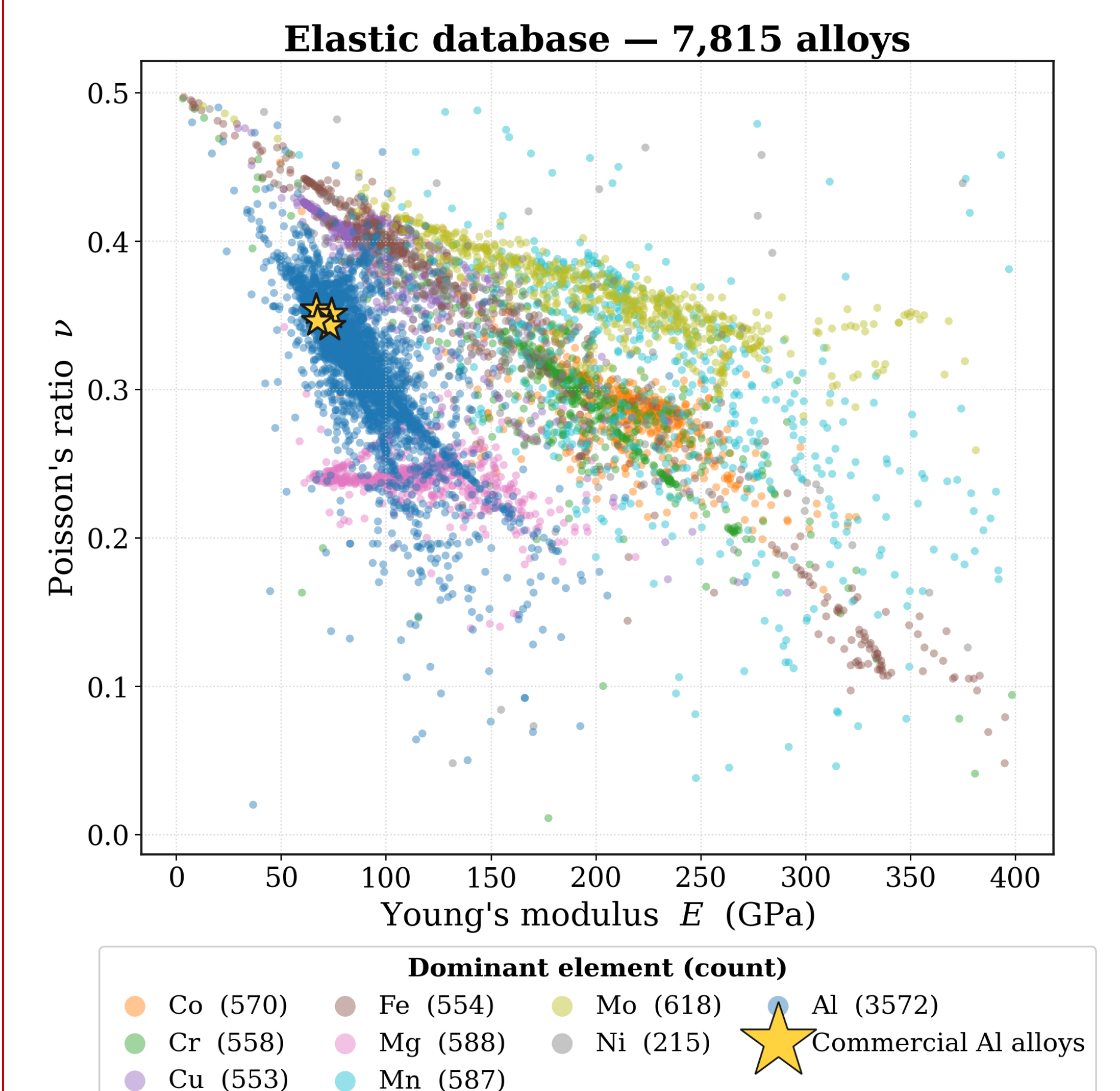


Fig. 5. E - ν for 7815 alloys, coloured by dominant element; commercial Al alloys starred.

COMPOSITION SURROGATE

$R_E^2 = 0.77$, $R_\nu^2 = 0.52$ on held-out validation. For Al₇₀Fe₃₀ the surrogate gives $E = 75.5$ GPa, $\nu = 0.334$ vs published 71–72 GPa, $\nu \approx 0.33$ — within $\sim 5\%$.

OPTIMIZED POTENTIAL

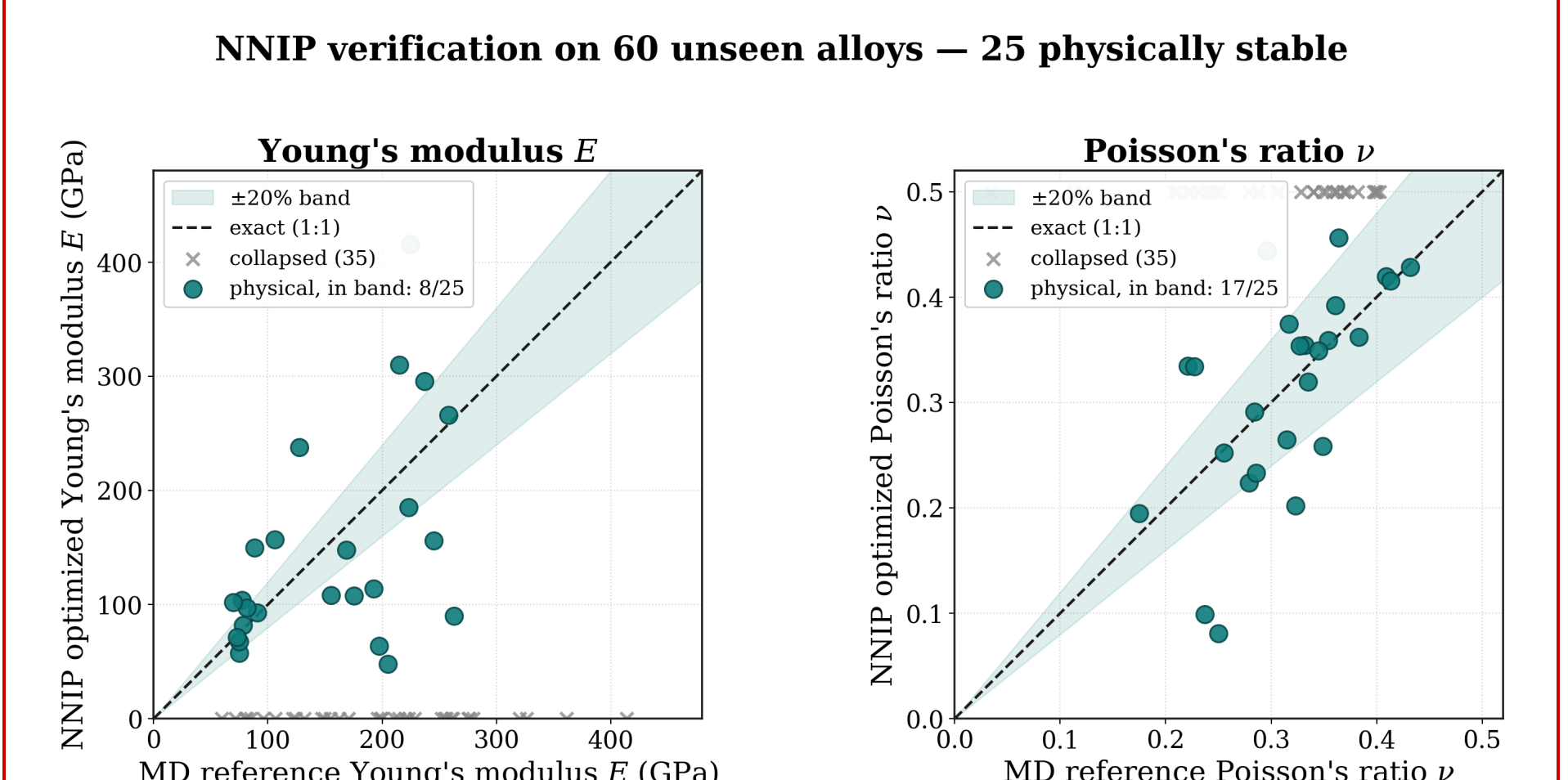


Fig. 6. Optimized potential vs MD on 60 unseen alloys: inside the $\pm 20\%$ band = match, $\times$ = collapsed (unphysical).

Proof of concept. Of 60 unseen alloys, 25 stayed physically stable; 17 matched ν and 8 matched E within $\sim 20\%$.

VALIDATION VS PUBLISHED DATA

Family	N	E MAPE (%)		ν MAPE (%)	
		ML	MEAM	ML	MEAM
Aluminium	33	2.3	7.4	4.0	4.5
Titanium	6	11.6	140.0	13.2	14.6
Stainless	3	69.6	324.1	75.5	47.3
Overall	43	10.8	42.2	7.2	7.1

Table 1. Surrogate (ML) and DFT-initialised potential (MEAM) vs published elastic data. Accuracy peaks where the corpus is densest (aluminium); sparse families inherit its bias — a data limit, not a method limit.

CONCLUSIONS & OUTLOOK

- **Automated end-to-end 12-element MEAM pipeline** — random compositions to an optimized potential, no hand fitting.
- **Al-alloy elastic prediction** from composition alone, within a few percent of published values.
- **DFT-seeded, NN-trained MEAM** — first-principles references refit through a neural surrogate.
- Next: more data, longer training, denser DFT references.